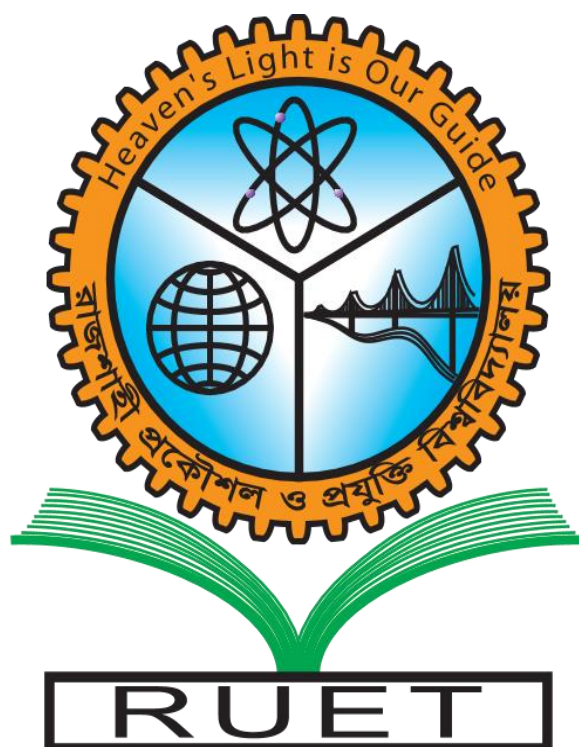


Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY



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REPORT ON: Smart Bin for Efficient Sorting of Recyclable, Non-Recyclable, and Compostable Materials Using YOLO Model.

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Submitted to	Submitted by
Md. Abu Hanif Pramanik Assistant Professor, Department of Electrical & Computer Engineering, RUET.	Maaz Ahmed 2010032 Puja Roy Dipa 2010040 Session 2020-2021 Department of Electrical & Computer Engineering, RUET.

Smart Bin for Efficient Sorting of Recyclable, Non-Recyclable, and Compostable Materials Using YOLO Model

I. INTRODUCTION

With increased economic growth and a rise in the standard of living in this modern world, so has the rate of solid waste risen to unimaginable rates, especially in urban settings. It is estimated that global wastes will reach 46 billion tonnes by 2050, while municipal solid waste will grow by 26%-45%, one-third of which is not collected, and much of what is collected is also being inadequately managed, with 42% openly dumped or burned [1]. This situation, therefore, points out with high concern the dire need for an effective waste management system especially in low and middle-income countries where it is highly problematic and inefficient [2]. Sorting of waste accordingly can contribute to effective management by reducing waste sent to landfills and improving the rates of recycling and composting, thus eventually leading to less environmental pollution.

This project presents an innovative approach by designing a camera-based smart automated bin powered by YOLOv8s, a deep learning model, for the efficient sorting of waste into recyclable, non-recyclable and compostable materials. The merged technology is not just a waste sorter, it also provides a facility to compost directly in the same bin. The proposed intelligent dustbin takes advantage of various classes such as electronic waste, glass, metal, plastic, organic materials and paper. The technology uses camera integrated Raspberry Pi 4 and computational capabilities to replace and improve the traditional method of sorting waste along with enhancing compost production. The main goal of this novel work is to educate environmentalism and create an automated waste management system.

The key contributions of this paper are as follows:

1. Design of an innovative automated smart bin that distinguishes waste into three main parts: recyclable, non-recyclable, and compostable materials as well producing compost within its unit.
2. A novel deep convolutional neural network based on YOLOv8 was used for effective waste segregation, thereby improving both detection and sorting efficiency and ensuring indisputable classification.
3. The dataset is quite diverse because it encompasses six categories of waste, namely electronic waste, glass, metal, plastic, organic material, and paper.
4. Real-time detection of the automated smart bin is the highlight and a very crucial part of the whole process when it comes to managing waste.

II. METHODOLOGY

This section describes the design and full implementation of the proposed smart waste-sorting system ranging from data collection to preprocessing, model training, performance evaluation, and hardware implementation using Raspberry Pi 4. The workflow puts deep learning into use, particularly the YOLOv8 model, which is very effective in various fields, from image classification to medical application [3]. It is used in this study for classifying and sorting waste in real time.

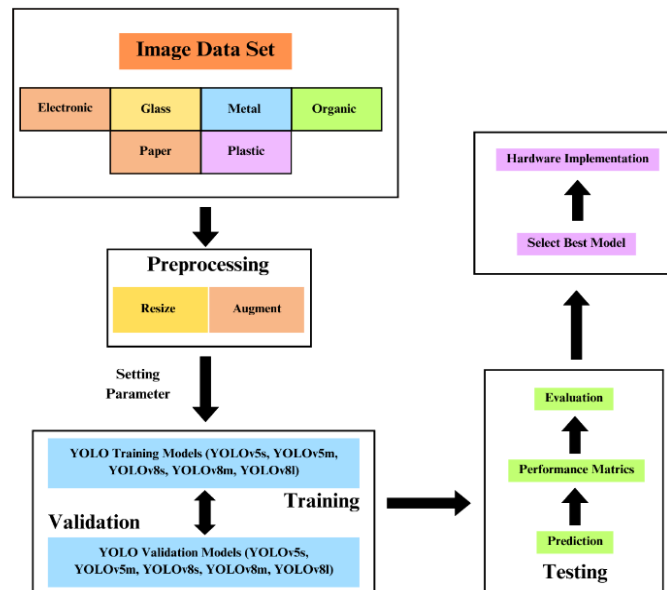


Figure 1. Methodology Workflow

A. Dataset Collection

The dataset used in waste detection and classification study was collected from multiple sources, such as a public dataset from Roboflow [4] which is an open-source platform of various kinds of dataset along with computer vision models also with custom-collected images which were used for training the models. The dataset consists of six different categories of waste material such as electronic, glass, metal, organic, paper, plastic. The custom images of the dataset used in this study were generated by using a camera and captured from different angles for each class. The total number of images was 2,340, and there were fourteen null images which were not classified under those waste material categories.

B. Preprocessing and Annotation

Preprocessing was done by adjusting the orientation and using Contrast Stretching. Horizontal and vertical flip augmentation was applied here, and this technique enhances the variability of the dataset hence increases the performance of detection and classification of different classes. The dataset used in this study consists of an average of 1.2 annotations per image. The annotations are classified into six distinct classes, which represent the six different classes of waste materials. The dataset comprises images of 512*384 pixels with a wide aspect ratio. This ratio is maintained across all images used in this dataset, which is an ideal ratio for images of this dataset for waste detection and classification. The average size of images used in this dataset is 0.20 megapixels. This size of the images varies from 0.02 megapixels to 16.93 megapixels, which mainly covers a wide range of image resolutions. The annotation process was carefully done by using Roboflow. Proper labeling was verified because it is a foundation to the success of YOLO models. This dataset was divided into three categories: train, test, validation. In the train set, there were 1,976 images which covered 84% images of the dataset and 182 images for both test and validation set which covered basically 8% images of the whole dataset. It was an 84-8-8 split. This dataset was used to train and evaluate all the YOLO models to determine their performance on waste detection and waste classification. Table-I covers the details of the dataset.

Table I: Dataset Distribution Based on Classes

Class	Electro-nic	Glass	Metal	Organic	Paper	Plastic
Train	341	327	341	320	327	320
Test	30	30	30	30	30	30
Valid	31	30	29	30	30	30

C. Model Training

Multiple variants of YOLOv8 and YOLOv5 were trained with the dataset, which include YOLOv8s, YOLOv8m, YOLOv8l, YOLOv5s, and YOLOv5m. Their performances were compared to determine the best model. Selected models underwent additional performance evaluations for further validation in ensuring accuracy and reliability. YOLOv8s was chosen for real time object detection because among all the trained models, YOLOv8s had the lowest inference time making it well suited for the task.

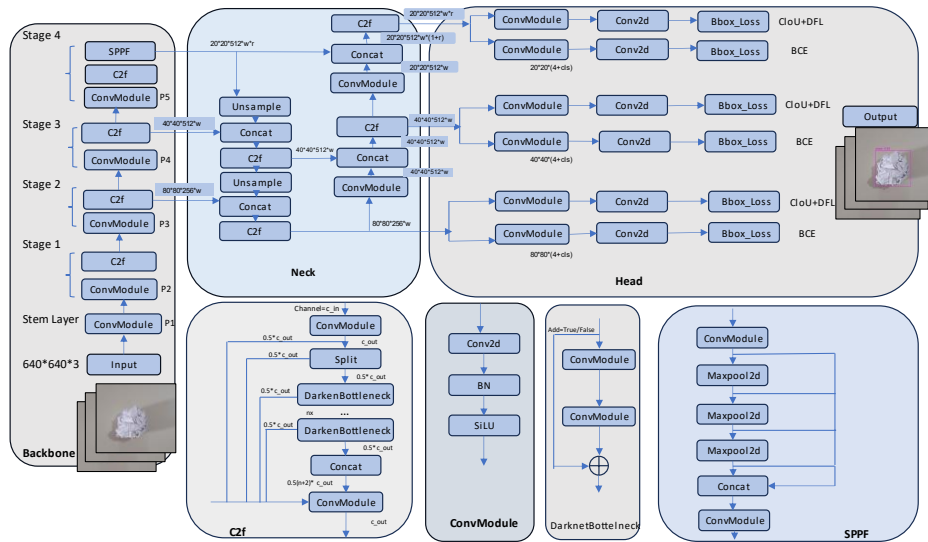


Figure 2. YOLOv8 Architecture

Figure-2 illustrates the architecture of YOLOv8. It consists of three important components: Backbone, Neck, and Head. The key features will be extracted by the Backbone from input images, the Neck will combine the features at different scales, and the Head will predict object detection. Training was performed on 50 epochs.

D. Model Evaluation

Standard object detection metrics were used to evaluate the trained models: mean Average Precision-mAP, Precision, Recall, F1-score. These provided insight into the models' accuracy regarding waste detection and classification. Loss curves were observed during training to avoid overfitting, and the best model was selected based on its performance on the validation set. Confusion matrices were also created to make sure the models performed the classification well across all different categories of waste identifying any misclassifications and ensuring that the model generalizes well across all waste types. The chosen model, therefore, was one that had a very high accuracy with low loss during testing and was suitable for real-time deployment.

III. MODEL RESULTS AND DISCUSSION

A. Results

In this study, precision, recall, F1-score, and inference time were computed and compared for five different YOLO models: YOLOv5s, YOLOv5m, YOLOv8s, YOLOv8m, and YOLOv8l. These metrics will then provide the basis of how each model performs in terms of object detection regarding both accuracy and speed-two very important factors in real-time detection systems.

Table II: Performance Evaluation for Different YOLO Models

Model	Precision (P)	Recall (R)	F1-Score	mAP@50
YOLOv5s	0.873	0.814	0.847	0.91
YOLOv5m	0.886	0.873	0.879	0.926
YOLOv8s	0.889	0.866	0.877	0.927
YOLOv8m	0.871	0.868	0.869	0.929
YOLOv8l	0.899	0.863	0.880	0.934

Table-II presents a comparison of precision, recall, F1-score and mean average precision at 50% intersection over union (mAP@50) for different YOLO models. The models were evaluated on a dataset with six classes, including electronic, glass, metal, organic, paper, plastic.

Among these, YOLOv8l had the highest accuracy, hence was the best in recognizing objects with the highest score of 0.899. At a close range, YOLOv8s is also performing remarkably well with a precision of 0.889, depicting excellent performance for accurate object detection. On the other hand, the YOLOv5m showed a precision of 0.886, while the YOLOv5s recorded a little lower at 0.873. Most notably, YOLOv8m had the lowest precision at 0.871.

In recall, YOLOv5m was outstanding because it captured most of the true positives with a score of 0.873. This will make it reliable in scenarios where identifying all relevant objects is critical. Behind this, the YOLOv8m and YOLOv8s had recall scores of 0.868 and 0.866, respectively. This is an indication that they are very good at picking true positives. The YOLOv8l had moderately good recall score compared to the others while YOLOv5s trailed with a much lower recall score of 0.814, indicating that for this model, it may miss a lot of actual positive detections.

Among these, the F1-score is balanced for precision and recall. The best overall performance, with an F1-score of 0.880, was given by YOLOv8l; this was seconded by YOLOv5m at an F1-score of 0.879, a model that featured very well-rounded performance. Next came YOLOv8s at 0.877, reflecting performance that was fair yet still lagging behind its bigger cousins. YOLOv8m reached 0.869, while YOLOv5s ranked last in F1-score at 0.847.

The last metric, mAP@50, shows that YOLOv8l is not only good at precision and recall but also provides the overall leading detection accuracy with a score of 0.934, making it versatile and powerful. Also, YOLOv8m and YOLOv8s had very good performances with scores of 0.929 and 0.927, respectively, showing the overall strength of the YOLOv8 series. By contrast, the scores of YOLOv5m and YOLOv5s were 0.926 and 0.910, respectively, demonstrating somewhat lower accuracy but still quite substantial.

Table III: Preprocessing, Postprocessing & Inference time Inference Time Comparison of Different YOLO Models

Model	Preprocessing Time(ms)	Inference Time (ms)	Postprocessing Time (ms)
YOLOv5s	0.2	1.9	0.8
YOLOv5m	0.3	5.8	1.1
YOLOv8s	0.4	2.1	0.7
YOLOv8m	0.3	3.9	0.8
YOLOv8l	0.5	5.2	1.1

Table-III represents the preprocessing time, post processing time and the inference time of all five YOLO models that were trained in this study. Regarding preprocessing time, YOLOv5s was the fastest, doing it in just 0.2 ms. In real-time use, this is somewhat efficient compared to other models such as YOLOv5m, which takes 0.3 ms. The pre-processing times for YOLOv8m are approximately 0.3 ms, while that of YOLOv8s is about 0.4 ms. YOLOv8l has the largest pre-processing time, with a score of 0.5 ms. As for inference time, YOLOv5s has the fastest processing at 1.9 ms, which is very suitable for real-time applications.

YOLOv8s follow with an inference time of 2.1 ms, also capable of fast detection. On the other side, the YOLOv8m takes 3.9 ms for inference, while the YOLOv8l is longer, taking 5.2 ms, testifying that the biggest models can already be quite resource-intensive and hence slow in real-life applications. The highest among them all is that of the YOLOv5m at 5.8 ms. As for the time of postprocessing, YOLOv5s shows once more how efficient it is at 0.8 ms, and its immediate successor, YOLOv8m, is almost at par in efficiency with 0.8 ms. YOLOv8s performed a little better with the time of 0.7 ms for postprocessing. YOLOv5m and YOLOv8l have the longest postprocessing time with 1.1 ms each, and this may have impacts on the overall speed when these models are integrated into an application. Among these, YOLOv8s turns out to be one of the best models for applications where real-time speed is vital along with accuracy. It has an ideal combination of fast processing speed combined with reliable detection. In contrast, models that run faster, like YOLOv5s, are significantly lagging behind the former in terms of accuracy and hence may prove less suitable in applications that require a high order of precision. Larger models, like YOLOv8l, are more accurate but at the cost of slower speed, probably not fitting for scenarios where speed is a must.

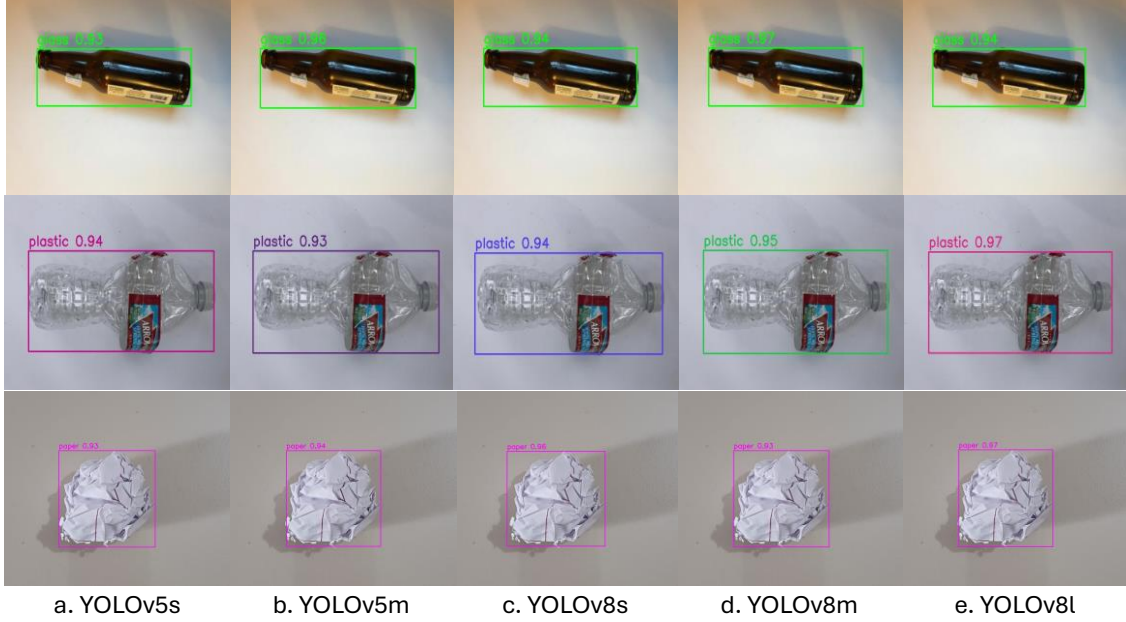


Figure 3. Individual YOLO models' performance. Predictions from YOLOv5 models with input image are shown for (A) YOLOv5s & (B) YOLOv5m. Predictions from YOLOv8 models are shown for (C) YOLOv8s, (D) YOLOv8m & (E) YOLOv8l

Along with all these metrics, confidence scores in object detection tasks remain among the key quantitative measures of how sure a model may be for any set of predictions being made. Figure.3 illustrates the object detection results with experimented YOLO models, where each model has detected a bounding box around the object of interest and includes the corresponding confidence score around the box. YOLOv8l tends to have a very high confidence score for all detected objects in view. In this respect, YOLOv8m also provides a strong performance with high confidence scores but slightly lower than YOLOv8l. YOLOv8s shows efficient detection but with slightly lower confidence scores than its bigger versions, specifically in the case of plastic. YOLOv5m yields decent results while YOLOv5s did comparatively poor and resulted in the lowest confidence scores and less reliable detection. Higher confidence scores imply a more reliable detection but other metrics such as precision, recall, F1-score, and inference time need to be taken into consideration in order to derive a full picture of how a model is able to perform. Not only does YOLOv8s show great accuracy, but it also shows high confidence scores with low inference time, making it very well-suited for any situation where speed and accuracy at detection are concerned.

B. Detailed Performance Analysis of YOLOv8s

Next the precision, recall, and mAP@50, of the YOLOv8s model in detecting six classes of waste are discussed to further evaluate its performance.

Table IV: Class-wise Performance Evaluation of YOLOv8s Model

Class	Images	Instances	Precision (P)	Recall (R)	mAP @50
All	182	202	0.889	0.866	0.927
Electronic	182	37	0.968	0.827	0.970
Glass	182	30	0.859	0.833	0.912
Metal	182	29	0.958	0.788	0.946
Organic	182	45	0.783	0.804	0.825
Paper	182	31	0.968	0.978	0.994
Plastic	182	30	0.795	0.967	0.975

Table-IV shows how well object detection works for types of waste materials like electronic, glass, metal, organic, paper and plastic. Precision (P) recall (R) and mean average precision at 50% intersection over union (mAP@50) are used as the matrices to evaluate the YOLOv8s model. The YOLOv8s model performs impressively with a mAP@50 of 92.7%, precision of 88.9%. Recall of 86.6%, for all classes considered together.

The paper category demonstrated outstanding performance, with a precision level of 96.8% and recall rate of 97.8% culminating a mAP@50 score of 99.4%. Likewise, the electronic and metal classes also displayed precision rates of 96.8 % and 95.8% respectively and corresponding mAP@50 metrics measuring at 97% and 94.6%.

The accuracy scores were slightly lower for plastic and organic categories at 79.5% and 78.3%. Their respective mAP50 scores were 97.5% and 82.5%. This indicates that although the model excels in areas of material detection tasks overall, there are still difficulties when it comes to identifying materials such as organic matters and plastic wastes.

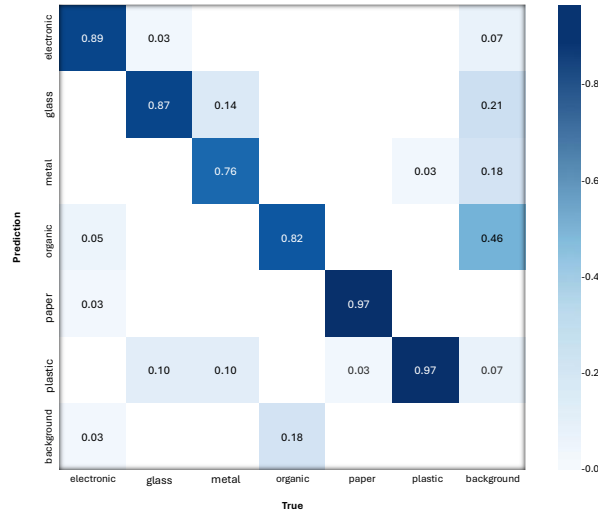


Figure 4. Normalized Confusion Matrix of YOLOv8s

In Figure-4, The normalized confusion matrix of YOLOv8s shows strong diagonal dominance, which indicates sharp and precise classification of target objects. On the contrary, several noticeable off-diagonal values indicate areas where the model struggles to differentiate some classes and could be a possible area for improvement.

Figure-5 illustrates a bar chart, highlighting instance counts of different waste categories, including organic, with the highest frequency, and metal as the lowest. Next to it in Figure-6, the bounding box visualization depicts the pattern of detection across different object sizes and locations. The following scatter plots in the figure investigate the object positions in the spatial distribution, according to x, y, width, and height, which show concentration points around the center. Moreover, the correlation of width and height is clustered, suggesting the same size of object detection in a cluster. Figure-7 is an F1-confidence curve showing the performance of YOLOv8s across the waste types, hence giving a higher F1-score with confidence scores between 0.5 and 0.9 in detecting paper. In general, performance for all classes is solid; the highest F1 score reaches 0.87 at 0.507 confidence. These figures point towards very good detection accuracy and spatial consistency throughout the dataset.

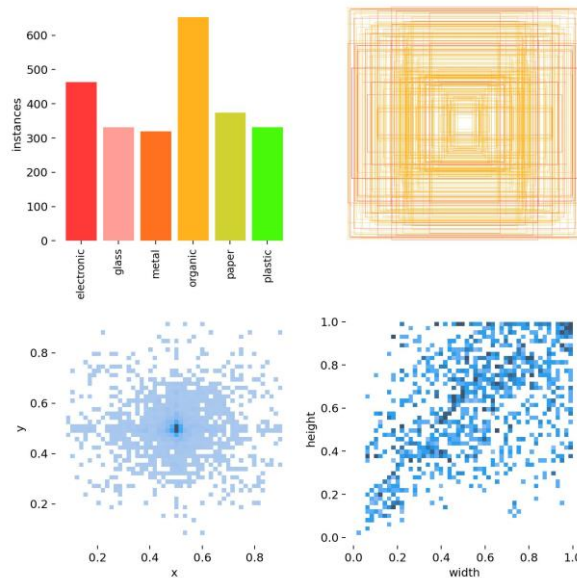


Figure 5. Visual Analysis of Waste Categories and Spatial Relationships of YOLOv8s

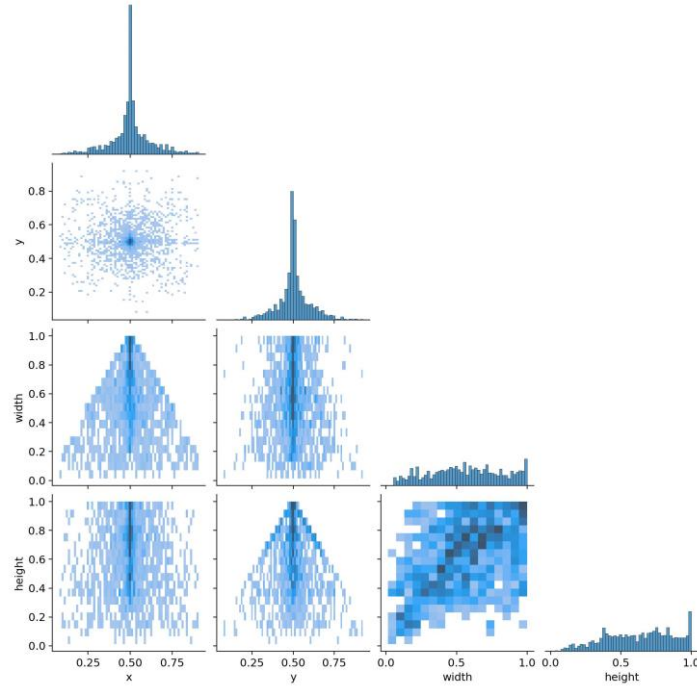


Figure 6. Histogram Representation of YOLOv8s

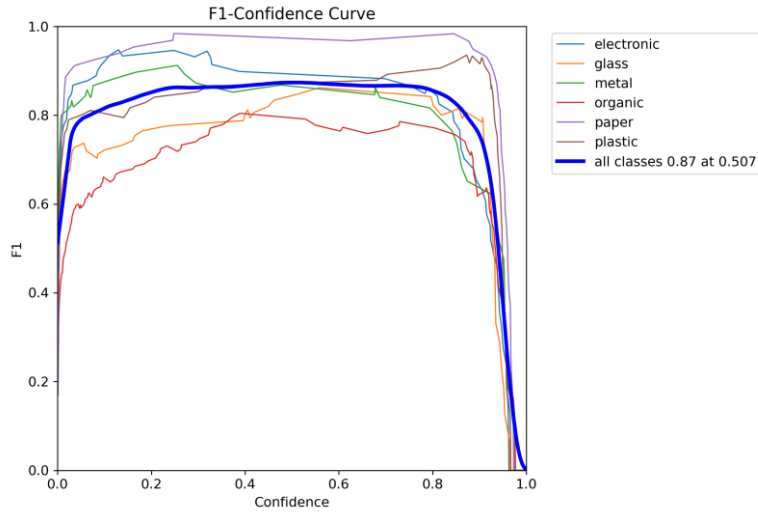


Figure 7. F1-Curve of YOLOv8s

C. Comparison with recent works:

Table-V compares the performance of the proposed model, YOLOv8s, against several existing models in waste detection. The proposed model outperforms the listed models with a precision of 0.889, recall of 0.866, F1 score of 0.877, and mAP@50 of 0.927 while the models such as YOLOv3 and YOLOv4 present lower metric values, proving the effectiveness of the proposed model to improve the detection accuracy and efficiency in performing the waste classification task. This comparison underlines the developments that have taken place with the use of the most recent YOLO architecture.

Table V: Performance Comparison of YOLOv8s with Existing Models

Reference	Model	Precision	Recall	F1 Score	mAP @50
Saurav Kumar et al. [5]	YOLOv3	---	0.87	---	0.9499
Minh Nguyen et al. [6]	YOLOv4	---	---	0.86	0.9027
M. Guru Vimal Kumar et al. [7]	YOLOv7	0.735	0.52	0.609	0.588
Meet Shroff et al. [8]	YOLOv8s	0.537	0.476	0.505	0.52
Pan Li et al. [9]	Enhanced YOLOv8s	0.774	0.626	0.692	0.709
Shenlin Liu et al. [10]	EcoDetect-YOLOv5	0.745	0.536	0.623	0.581
Proposed Model	YOLOv8s	0.889	0.866	0.877	0.927

D. Hardware Simulation

The design of the smart bin comprises three different compartments, each for a specific waste type. A machine learning-based design was proposed for devising an eco-friendly solution for efficient sorting of waste. The integrated object detection algorithm of YOLOv8s does the auto-classification and identification of different kinds of waste, such as recyclable ones (Electronics, Glass, Metal, Paper, Plastic), compostable ones (organic), and non-recyclable ones. For detecting waste that goes into the non-recyclable materials, a motion detector was used such that it detects whether any item was put on the surface of the bin, and in case it is not detected under any class, then after a certain amount of time (4s) it will be put under a non-recyclable item. At the top of the dustbin, an extension is given for housing the camera. The camera monitors the area where the waste will initially fall. Once the trash is in that area, the camera detects the object. The Raspberry Pi 4 controls actuators and servos in sorting out the trash physically, opening the correct compartment according to the classification. Immediately after the waste is placed, the funnel mechanically starts rotating with the help of a motor, and it rotates at a predefined angle based on the waste type. Once this rotation is achieved, it acts as a trigger for opening the top surface, directing the deposited waste into the concerned chamber. Hence, sorting of the waste is ensured. In the compostable chamber, the organic waste is further processed. There is a stirrer inside it that keeps on agitating the material constantly, thus facilitating compost production. The system is designed to work autonomously, with the following key components:

- **Camera Module:** It captures an image of the waste put in the bin.
- **Sensors:** The motion sensors detect the deposition of the waste and send signals to the camera to capture the image.
- **Actuators:** Motors, based on the classification provided by the YOLOv8s model, actuate and separate waste into the designated compartment; be it recyclable, non-recyclable, or compostable.
- **Composting:** Even the compostable compartment has a mechanism for composting where organic waste could begin decomposing to create compost automatically.

This whole system is designed to be very energy-efficient, cost-effective, and scalable for both residential and community-level applications. The whole hardware was designed and simulated virtually. It works properly as intended. The design and simulation parts are attached below in Figure-8.

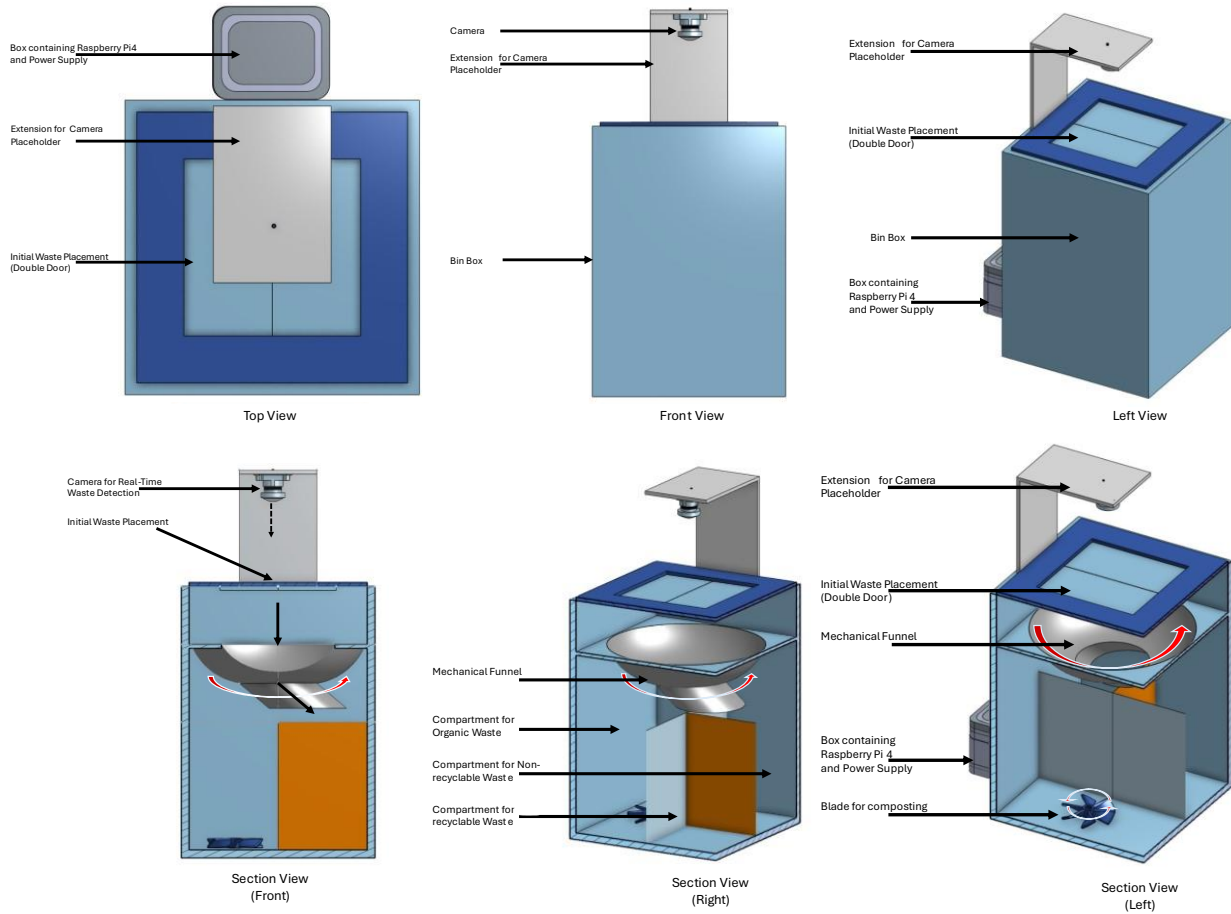


Figure 8. 3D Structure of The Smart Bin

E. Discussion and Analysis

According to precision, recall, and F1-score analysis, YOLOv8l is leading the chart with the best precision of 89.9% and an F1-score of 88%. In other words, YOLOv8l is the most precise in this group. However, its inference time is 5.2 milliseconds,

which is unsatisfactory for an application that requires real-time processing speed. YOLOv8s, on the other hand, offers a faster inference time of 2.1 milliseconds with a mAP@50 of 92.7% and an F1-score of 87.7%, making it the better choice for real-time tasks requiring both speed and accuracy. In summary, YOLOv8l is ideal when higher precision is prioritized over speed, but for real-time detection tasks, YOLOv8s strikes the best balance between accuracy and low latency. Given this work's focus on real-time detection, YOLOv8s is proposed as the most suitable model. Implementing on hardware, it shall demonstrate satisfactory detection speed and accuracy, making it a strong candidate for real-world, fast, and reliable object detection applications.

IV. CONCLUSION

This study addresses the advantages of real-time detection using YOLO models through bringing light on the trade-offs between speed and accuracy. Though YOLOv5 models were good at real-time object detection, YOLOv8 emerged as a promising framework for faster detection tasks, specially YOLOv8s, which offered a well-balanced solution for applications requiring both fast inference time and high-end performance. Although it did not achieve the highest precision or recall compared to YOLOv8l, it did achieve a satisfactory result as it had fastest inference time among the trained models. So, it is the best suited model to deploy in resource-limited environments, such as on Raspberry Pi 4, showcasing its potential for real-world usage. Despite the extraordinary result, it should be noted that there are some limitations. In critical settings prioritizing real time speed might slightly affect the accuracy of detection efforts, enhancing the robustness of real-time detection systems. Also, customized YOLO models may produce much more accurate results. Furthermore, large and more diverse datasets can make implementations more acceptable in real life scenarios. Overall, the research adds to the increasing knowledge base regarding real-time detection systems.

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